Problem Description | Solution | Video | Comments 1 | Playground

Transpose of a Matrix

Easy | Linear Algebra

Write a Python function that computes the transpose of a given 2D matrix. The transpose of a matrix is formed by turning its rows into columns and columns into rows. For an mÃ‰n matrix, the transpose will be an nÃ‰m matrix.

Example:

Input:

```
a = [[1, 2, 3], [4, 5, 6]]
```

Output:

```
[[1, 4], [2, 5], [3, 6]]
```

Reasoning:

The input is a 2×3 matrix. The transpose swaps rows and columns: the first row [1, 2, 3] becomes the first column, and the second row [4, 5, 6] becomes the second column, resulting in a 3×2 matrix.

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Contributors:

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Notebook Mode | NumPy

```python
# This needs to be imported at the top of the file, inside the Deep-ML code writing section.
import numpy as np


def transpose_matrix(a: list[list[int | float]]) -> list[list[int | float]]:
    """
    Transpose a 2D matrix by swapping rows and columns.
    Args:
        a: A 2D matrix of shape (m, n)
    Returns:
        The transposed matrix of shape (n, m)
    """

    # First, converting the python list of lists into a numpy array.
    a = np.array(a)

    # Then transposing the matrix using the numpy function 'transpose'.
    a_transposed = np.transpose(a)

    # Finally, converting the transposed numpy array into a python list of lists.
    a_transposed = a_transposed.tolist()
```

Run Code | Submit Code | Reset | Save Code | Submissions 1

Case 1 +

Test Code

```
print(transpose_matrix([[1, 2], [3, 4], [5, 6]]))
```

Expected Output

```
[[1, 3, 5], [2, 4, 6]]
```

Test Results 5/5

Test 1 | Test 2 | Test 3 | Test 4 | Test 5

Test Case

```
print(transpose_matrix([[1, 2], [3, 4], [5, 6]]))
```

Expected

```
[[1, 3, 5], [2, 4, 6]]
```

Your Output — Passed

```
[[1, 3, 5], [2, 4, 6]]
```

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